Problem B

Predicting Penalty Trajectories: What Two Early Positions Can Resolve

Abstract

We use the ball's positions 50 and 100 ms after release to estimate the probability of direct entry and the most likely point of entry on the goal plane. For the prescribed kick, assuming exact release timing, centimetre rounding and horizontal wind speeds up to 3 m/s, the most probable cell is centred at $(Y, Z) = (2.56, 2.07)$ m. Six cells covering 0.3572 m^2 contain 90.52% of the model probability conditional on direct entry. The median crossing time is 0.399 s. Two admissible flights reproduce the same readings and cross 8.76 cm apart. For this pair the worst-case error of any point forecast is at least about 4.38 cm.

We calibrated prediction regions on 90 synthetic kicks and tested them on 120 others, of which 71 entered directly. With rounding alone, the two-frame and late-frame aerodynamic forecasts both covered 70 of 71 entries at their 95% calibration targets. Their regions averaged 0.585 and 3.482 m^2, respectively. A two-frame gravity model covered the same number in 1.222 m^2. Adding the 50 ms reading reduced mean point error by 3.80 cm relative to the late frame alone. With 6 mm independent and 8 mm shared tracking errors, gravity and two-frame aerodynamics both covered 67 of 71, at calibration targets of 80% and 95%, using 0.933 and 1.385 m^2, respectively. In an independent test with tracking errors and a pooled 90% target, the two-frame aerodynamic regions covered 343 of 368 direct entries overall, including 90 of 110 with stable spin. The three available match records lacked an independent release time, so accuracy on real penalties remains open.

1 What the first 100 ms reveal about the flight

At 100 ms the ball in the prescribed penalty has travelled only 2.96 m toward the goal. The two readings reveal its average velocity over the preceding 50 ms, yet the crossing depends on how drag and transverse forces act over the remaining 8.04 m (Fig. 1). The task is to predict where the ball is most likely to enter the goal, estimate the uncertainty in that location and test the method [1].

We first ask which forces can explain the early positions. We then follow the admissible flights to their first event. This gives the probability of direct entry and, among flights that enter, a point and joint region on the goal plane (Table 1). The problem supplies no final position for the prescribed kick, so we assess accuracy on separate kicks.

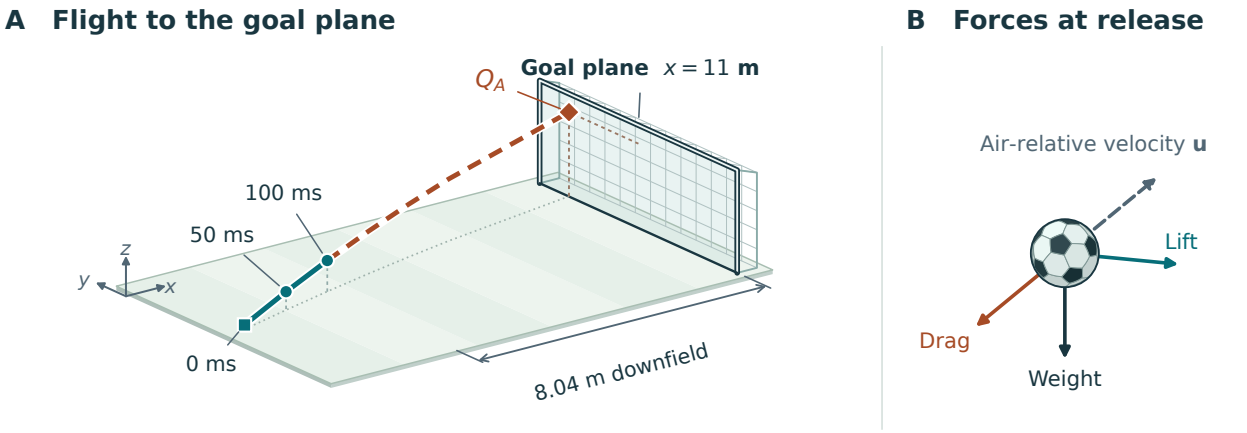


Figure 1. Two readings and one admissible flight. In A, the square marks release and the circles mark the prescribed ball centres, given below in (x, y, z) order. Dashed segments join archived positions of flight A to its crossing Q_A (Sec. 4). The dotted line on the turf is the ground projection. B shows the same flight at release in still air. Drag opposes the air-relative velocity, lift acts perpendicular to it in three dimensions, and weight points down. Arrow lengths are schematic.

Table 1. How the two observations available by 100 ms enter each part of the problem.

Task	Forecast or test	Evidence
Develop a predictor	First-event probabilities and highest-mass goal cell	Eqs. (3), (5)
Estimate uncertainty	Joint region and sensitivity to observation and force assumptions	Figs. 3–5
Apply and test	Prescribed kick and synthetic kicks at the same 100 ms cutoff	Eq. (6), Fig. 6

1.1 Observations and possible outcomes

Let x point from the penalty spot toward the goal, y run across its face and z measure height above the ground. The ball centre starts at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m. Its two prescribed positions, in metres, are

$$\boldsymbol{r}(0.050\,\mathrm{s}) = (1.50, 0.45, 0.44), \qquad \boldsymbol{r}(0.100\,\mathrm{s}) = (2.96, 0.86, 0.74). \tag{1}$$

For the reference forecast we take the release position and observation times as exact and interpret the displayed coordinates as rounded to the nearest centimetre. Each later coordinate q then represents $[q-\epsilon, q+\epsilon)$, with $\epsilon = 0.005$ m. The problem supplies no tracking accuracy, ball type, wind or spin. Section 5 examines how different observation assumptions affect this kick.

We follow the ball to its first forward crossing of $x = 11$ m or its first descent to one ball radius above ground. A direct entry G occurs when the centre crosses the 7.32 m wide opening with $-3.66 < Y < 3.66$ m and $0 < Z < 2.44$ m. The other outcomes are a crossing outside the opening, earlier ground contact, or neither event by 1.25 s. We estimate all four probabilities and the distribution of $Q = (Y, Z)$ conditional on G. This ball-centre convention omits frame collisions, rebounds and goalkeeper interventions.

1.2 How much wind would drag alone require?

The mean horizontal velocities over the first and second 50 ms are $(30.0, 9.0)$ and $(29.2, 8.2)$ m/s. Their directions differ. Under drag alone in a constant horizontal wind $\boldsymbol{w}$, the horizontal velocity keeps its direction relative to the air. The horizontal displacements $\boldsymbol{p}_i$ from release must therefore satisfy

$$\det(\boldsymbol{p}_1 - t_1\boldsymbol{w},\, \boldsymbol{p}_2 - t_2\boldsymbol{w}) = 0, \qquad \|\boldsymbol{w}\| \geq \frac{|\det(\boldsymbol{p}_1, \boldsymbol{p}_2)|}{\|t_2\boldsymbol{p}_1 - t_1\boldsymbol{p}_2\|}. \tag{2}$$

Treating the displayed positions as exact gives a minimum wind speed of 14.849 m/s. For fixed wind the determinant, the signed area spanned by the two vectors, is linear in each coordinate separately. Its extrema on the closed rounding boxes therefore occur at their corners. Allowing for rounding lowers the infimum to 4.353 m/s, still above the 3 m/s wind cap. Under this cap the kick requires a transverse force, but the observations leave its source open. A wind that passes the collinearity test must still satisfy the full flight equations.

In still air the horizontal path under drag alone must be a straight line through the origin. For the positive coordinates in this example, the closure of the slope interval from frame i is

$$\left[\frac{y_i - \epsilon}{x_i + \epsilon},\, \frac{y_i + \epsilon}{x_i - \epsilon}\right].$$

The two ranges first touch at $\epsilon = 7.279$ mm. Thus a 5 mm per-coordinate bound resolves a departure from a straight horizontal path. At a 1 cm bound the readings admit such a path.

2 From aerodynamic forces to a crossing forecast

2.1 Three force laws and their parameter ranges

Write $\boldsymbol{v} = \dot{\boldsymbol{r}}$ for velocity over the ground, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ for velocity relative to the air, and $U = \|\boldsymbol{u}\|$. The three force models give accelerations

$$\begin{aligned} \boldsymbol{a}_{\mathrm{D}} &= \boldsymbol{g} - \kappa U \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{S}} &= \boldsymbol{a}_{\mathrm{D}} + \boldsymbol{\beta} \times \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{E}} &= \boldsymbol{g} - KC_D U \boldsymbol{u} + KC_L U^2 \boldsymbol{n}. \end{aligned} \tag{3}$$

Model D has quadratic drag. S adds a transverse force with constant coefficient vector $\boldsymbol{\beta}$. In E, $K = \rho\pi R^2/(2m)$ and lift acts along the unit vector $\boldsymbol{n}$ in the direction of $\boldsymbol{\omega} \times \boldsymbol{u}$. Lift vanishes when the cross product is zero. We use $g = 9.81$ m/s^2 downward, radius $R = 0.11$ m, mass $m = 0.43$ kg

and air density $\rho = 1.20$ kg/m^3. With inputs in SI units, the coefficients for E are

$$C_D = 0.155 + \frac{0.346}{1 + \exp[(U_c - 12.19)/1.309]} + \delta, \qquad U_c = \min(29.9, \max(9.3, U)),$$
$$C_L = 0.38 \tanh\Big(5R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2\Big). \tag{4}$$

The drag transition in E comes from a wind-tunnel fit for a nonspinning Teamgeist ball [2]. We clip its speed argument to 9.3–29.9 m/s, the initial-speed range of a separate 29-flight study [3]. Extrapolation to the unknown ball remains uncertain. The offset δ, the saturating lift curve and the broad spin range in E are modelling choices. In S, $\boldsymbol{\beta}$ is an effective transverse-force coefficient.

We assign D, S and E prior probabilities $0.2, 0.4, 0.4$ and draw wind uniformly from a horizontal disk of radius 3 m/s. Launch-velocity components are independently uniform over $[18, 38]$, $[-12, 12]$ and $[1.5, 11]$ m/s along x, y, z. For D and S, κ is uniform on $[0.006, 0.024]$ m^{-1}. The components of $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E are uniform within ± 0.8 s^{-1} and ± 65 rad/s, respectively. In E, δ is uniform on $[-0.04, 0.27]$.

2.2 Force scales over the remaining flight

The first displacement gives a speed scale $U_0 \simeq 32$ m/s. At $\kappa \simeq 0.020$ m^{-1}, the 0.4 s flight lasts about 0.26 of the drag time $(\kappa U_0)^{-1} \simeq 1.56$ s. A transverse coefficient of 0.5 s^{-1} gives an initial acceleration scale of 16 m/s^2. A residual transverse acceleration of 1 m/s^2 sustained over the remaining 0.30 s would accumulate about 4.5 cm of displacement.

Constant wind also gives an energy check. Because $\boldsymbol{u} \cdot (\boldsymbol{\beta} \times \boldsymbol{u}) = 0$ and $\boldsymbol{u} \cdot \boldsymbol{n} = 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{U^2}{2} = \boldsymbol{u} \cdot \boldsymbol{g} - d(U)U^2, \qquad d(U) \geq 0.$$

where $d(U)$ is the nonnegative drag rate. A transverse force turns the air-relative velocity, while drag dissipates air-relative kinetic energy. We use this identity to bound acceleration in the match records.

2.3 Weighting trajectories and mapping their crossings

For each draw of wind and aerodynamic parameters, we estimate launch velocity from the latest visible position under gravity and correct the estimate three times by integrating to that position. We draw velocities from a local uniform distribution centred on this estimate, with half-width $0.007\,\mathrm{m}/t$ per axis at observation time t, or from the original prior with probability 2%. Under rounding alone, a trajectory inside every observed interval receives the prior-to-proposal density ratio. All others receive zero weight. For parameters θ, proposal density q_M and likelihood L,

$$w_{Mj} = \frac{p(\theta_{Mj} \mid M) L(D \mid \theta_{Mj}, M)}{q_M(\theta_{Mj})}, \quad \widehat{Z}_M = \frac{1}{N_M}\sum_j w_{Mj}, \quad \widehat{p}(M \mid D) \propto p(M)\widehat{Z}_M. \tag{5}$$

Sampling nuisance parameters from their priors cancels those densities in the ratio. The correction for launch velocity remains. The mean weight $\widehat{Z}_M$ estimates the evidence for force model M. Systematic resampling limits propagation to 512 trajectories per model per test kick, or 8,192 per model per seed for the prescribed kick. Both prescribed-case runs stayed below their cap.

We use fourth-order Runge–Kutta steps of 25 ms through the two prescribed observations and at most 12.5 ms thereafter. Linear interpolation locates plane crossings and ground contact within a

step. Four numerical checks, including independent SciPy DOP853 integrations [4], agreed within 0.171 mm in event position and 0.038 ms in time.

First-event probabilities use the weights of all trajectories. For direct entries we normalize their weights on a 30×10 equal-area goal grid. The centre of the most probable cell gives the point forecast. Adding cells in decreasing order of probability until they contain at least 90% gives the raw joint region (Fig. 2). Zero total weight leaves the forecast undefined. If only the direct-entry weight is zero, we can still report event probabilities. We flag test kicks with an effective sample size $\mathrm{ESS} = 1/\sum_j \bar{w}_j^2$ below 25, using normalized weights including the model and timing mixtures. Low ESS indicates that few draws carry much of the weight. A finite sample can miss admissible flights altogether.

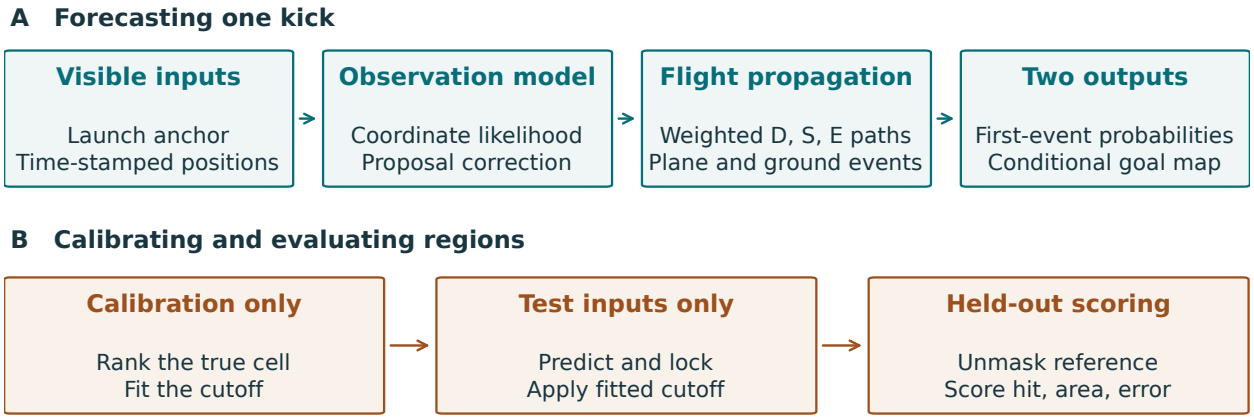


Figure 2. From early positions to a forecast. A follows one kick from its observed positions to first-event probabilities and a location map conditional on direct entry. B shows how we calibrate and test regions on separate kicks (Sec. 6). Only scoring uses the held-out reference crossings.

3 Where the prescribed kick is most likely to cross

3.1 Modal cell and joint prediction region

Conditional on direct entry, the most probable cell contains 29.14% of the probability (Fig. 3). Its centre, approximately $(Y, Z) = (2.56, 2.07)$ m, lies 1.098 m inside the positive-y post and 0.366 m below the crossbar. The six most probable cells form the rectangle

$$2.196 \le Y \le 2.928 \text{ m}, \qquad 1.708 \le Z \le 2.196 \text{ m}. \tag{6}$$

Together these cells occupy 0.3572 m^2 and contain 90.52% of the conditional probability. The median crossing time from release is 0.399 s, with 5th and 95th percentiles of 0.392 and 0.407 s.

Across two independent runs of 131,072 proposals per model per seed, all 3,469 trajectories with positive weight crossed inside the opening. Thus the estimated direct-entry probability is one under these assumptions and this finite sample. Posterior model weights were $0, 0.346, 0.654$ for D, S and E, consistent with the wind bound that excludes D in Eq. (2). The prescribed kick has no measured crossing or outcome against which to check these estimates.

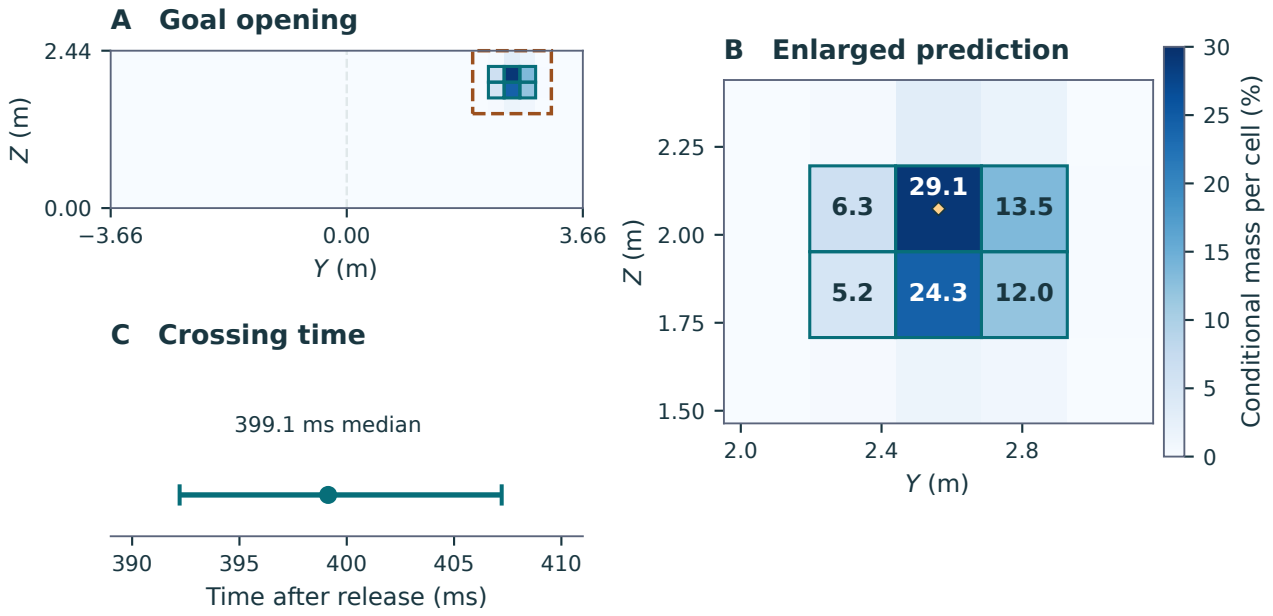


Figure 3. Predicted location and time of direct entry. A locates the enlarged area on the goal. B shows the probability per 0.244 m square cell using the same colour scale. The outline encloses the six most probable cells and the diamond marks the modal centre. C gives the 5th, 50th and 95th percentiles of the crossing time.

3.2 How precisely can the hotspot be located?

Both seeds selected the cell centred at $(2.562, 2.074)$ m. Varying E's share of the combined S/E prior from 10% to 90% also preserved this cell. A finer grid moved the modal centre to $(2.501, 2.013)$ m, and a half-cell shift of the original grid gave $(2.684, 1.952)$ m. The extra digits identify a cell on each grid. The six-cell region in Eq. (6) quantifies the nominal spatial uncertainty. The continuous conditional mean, $(2.603, 1.970)$ m, minimizes posterior squared distance. Our aerodynamic point forecasts use the centre of the cell with greatest probability at the stated resolution.

The force law also matters over the last eight metres. Fitting launch velocity by least squares under gravity alone gives $(3.217, 1.957)$ m. Fitting constant acceleration exactly to release and both frames gives $(2.495, 1.920)$ m.

4 A lower bound on two-position prediction error

4.1 Two flights consistent with the same observations

How closely can any point forecast fit every admissible flight? In model S we fixed the wind at zero and β_x at either -0.5 or $+0.5$ s^{-1}. For each choice we fitted the six coordinates at 50 and 100 ms by varying the three launch-velocity components and κ, β_y, β_z. Both fits lie within the prior ranges, with maximum coordinate residual below 5×10^{-13} m. An independent integration with MATLAB `ode45` reproduced both crossings [5].

Both drag coefficients round to 0.02007 m^{-1}. The fitted (β_y, β_z) pairs are $(-0.206, -0.463)$ and $(0.085, -0.251)$ s^{-1}, respectively. At any instant, a component of $\boldsymbol{\beta}$ along $\boldsymbol{u}$ contributes no transverse force. As gravity turns the velocity, that component can influence the later flight. Small adjustments in launch velocity allow different force histories to agree at both early frames.

The two flights yield identical centimetre readings but cross at $Q_A = (2.57378, 2.00251)$ m and

$Q_B = (2.62831, 1.93396)$ m, 8.76 cm apart. Any point predictor must give them the same $\widehat{Q}$. The triangle inequality then gives

$$\max\{\|\widehat{Q} - Q_A\|, \|\widehat{Q} - Q_B\|\} \;\geq\; \tfrac{1}{2}\|Q_A - Q_B\| \;\approx\; 0.0438 \text{ m}. \tag{7}$$

Thus at least one of the two errors reaches about 4.38 cm whenever both flights are admissible, regardless of their relative frequency.

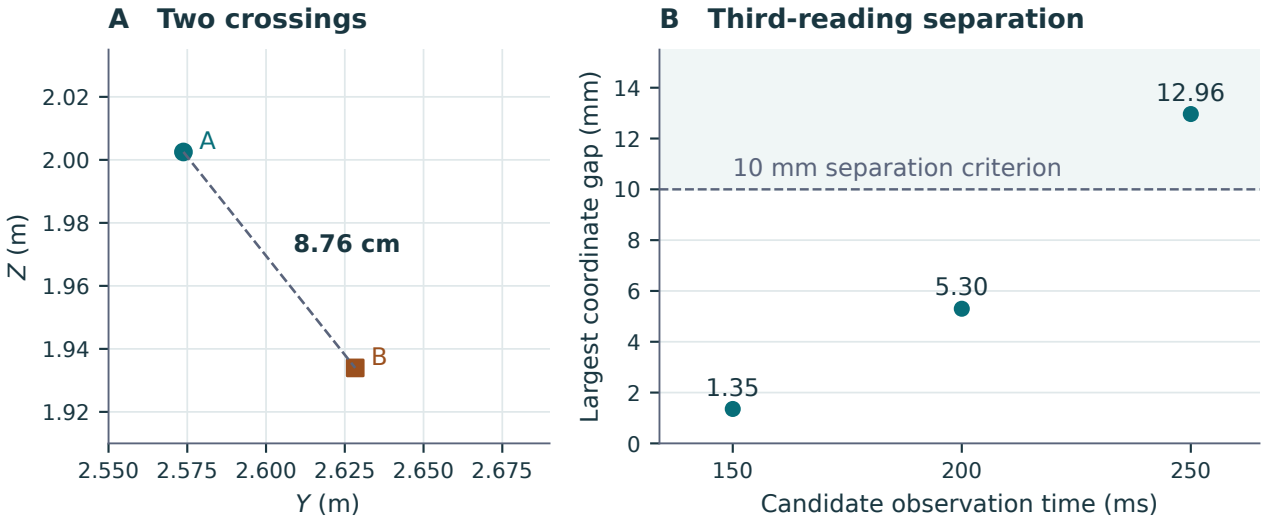


Figure 4. Two admissible flights separate after the second reading. A shows the two model-S crossings with the same scale along both axes. B gives the largest coordinate gap at three candidate observation times. With a 5 mm bound on each coordinate, the reading boxes separate once this gap exceeds 10 mm.

5 How measurement error affects the forecast

5.1 Position, timing and rounding errors

For axis a, let d_{ia} denote the position in frame i displayed to the nearest $q = 0.01$ m. We model the displayed value as

$$d_{ia} = \mathcal{R}_q[r_a(t_i + \tau; \theta, M) + b_a + e_{ia}], \quad b_a \sim N(0, \sigma_b^2), \quad e_{ia} \sim N(0, \sigma_e^2). \tag{8}$$

Here b_a is a bias common to the frames of one kick, e_{ia} is independent across frames and axes, and τ shifts all the frame times together. Conditional on b, τ, the probability of the displayed readings is

$$L = \prod_{i,a} \left[\Phi\!\left(\frac{d_{ia} + q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) - \Phi\!\left(\frac{d_{ia} - q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) \right]. \tag{9}$$

Here Φ is the standard normal cumulative distribution function. With rounding alone, L is the indicator that every coordinate lies in its interval. We integrate over b by sampling from its prior and, for a uniform timing offset on a bounded interval, over τ by five- or seven-node Gauss–Legendre quadrature with the same total proposal budget. The synthetic tests prescribe these error distributions.

A position error small enough to miss by eye can dominate the inferred acceleration. For an exact release coordinate and frame errors on one axis with common variance σ^2 and correlation ρ_e, the acceleration error inferred from the second difference at spacing h has variance

$$\operatorname{Var}\!\left(\frac{e_2 - 2e_1}{h^2} \right) = \frac{(5 - 4\rho_e)\sigma^2}{h^4}. \tag{10}$$

A common timing shift contributes $-v\tau/h^2$ to the acceleration inferred from uniform motion. At $v = 30$ m/s and $h = 50$ ms, a 1 ms shift produces an apparent acceleration of 12 m/s^2. A spatial bias cancels from the second difference if it affects release and both frames equally. If the release coordinate is exact, a bias shared only by the two frames survives and can resemble aerodynamic curvature.

5.2 Sensitivity of the prescribed forecast

We varied one assumption at a time in fourteen scenarios and refitted the same readings, using the nominal two seeds and 131,072 proposals per model per seed. Table 2 and Fig. 5 use displacement of the continuous conditional mean to track movement independently of the grid. Region areas use the original grid.

Table 2. Changes from the nominal forecast under individual assumptions. Shift is the distance between continuous conditional means. Area is the raw 90% region. Changing ball mass affects the force-to-mass factor in E. D and S use fitted acceleration coefficients.

Changed assumption	Mean shift (cm)	Area (m^2)
Nominal assumptions	0.00	0.357
Coordinate half-width: 10 mm	14.11	0.893
Both observation times: +1 ms	3.73	0.357
Wind cap: 0 m/s	0.83	0.357
S: halve β component limits	3.97	0.357
E: halve ω component limits	9.08	0.357
E: ball mass 410 g	0.25	0.357
E: ball mass 450 g	0.45	0.357
6 mm independent + 8 mm shared noise	23.70	1.072

Doubling the coordinate bound to 10 mm moves the mean by 14.1 cm and expands the region to 0.893 m^2. With 6 mm independent tracking error and 8 mm shared bias before rounding, the mean shifts 23.7 cm. On the same grid the most probable cell now lies at $(3.294, 1.830)$ m and the raw region covers 1.072 m^2. Removing wind shifts the mean by 0.83 cm for this kick. Halving E's spin range shifts it by 9.08 cm. The 410–450 g scenarios use the endpoints of the match-ball mass range [7] while holding radius fixed.

A tighter 2.5 mm coordinate bound leaves ESS 68, below the prescribed-case threshold of 200. Moving both observations 1 ms earlier leaves ESS 1. A 5 ms shift in either direction produces no positive-weight samples. These draws give too little support to locate the crossing reliably. Varying one factor at a time also leaves interactions among timing, noise, wind and spin unresolved.

6 Testing the two-frame forecast at 100 ms

6.1 Paired forecasts at a common deadline

At 100 ms both aerodynamic predictors know the latest ball position. We give one of them the 50 ms reading as well. Both start computing at 100 ms with the same priors, force laws and solver. Each draws 8,192 proposals per force model, propagates up to 512 positive-weight samples per model and reports the centre of the most probable goal cell. A gravity-only control uses both readings, predicts a continuous ballistic crossing and assigns direct entry a probability of zero or one.

In the first test (Z09) we generated 90 calibration kicks and 120 test kicks with fresh random seeds. The reference solver used a 1 ms step and different force laws from the predictor, with balanced blocks

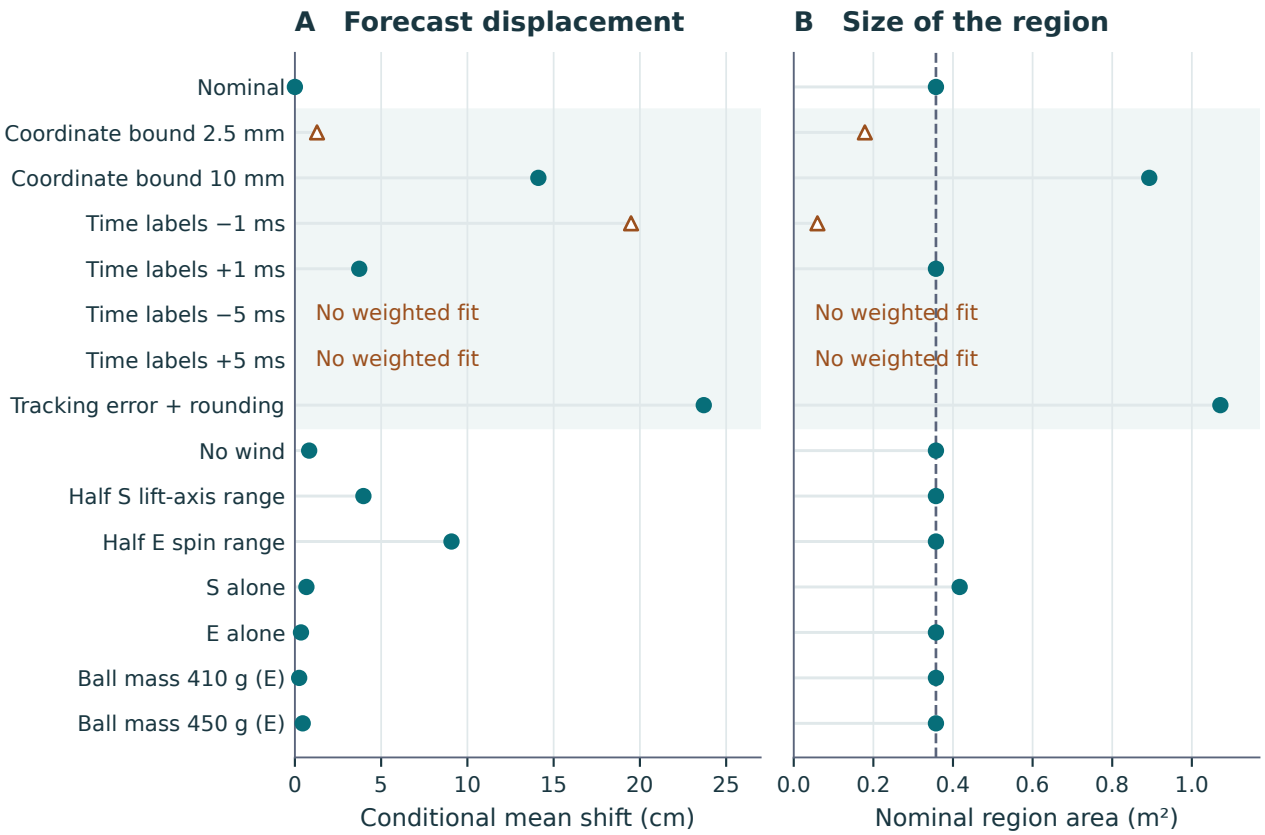


Figure 5. How the expected crossing and region size change. A shows displacement of the conditional mean. B shows the raw 90% region area and its nominal value (dashed). Shading identifies observation scenarios. Open triangles mark ESS below the prescribed-case threshold of 200. Rows with zero positive weight have no estimate.

of zero-spin, stable-spin and low-spin wake trajectories (Appendix D). We calibrated the regions on 55 direct entries and scored locations on the 71 test entries. All 120 test kicks contributed to the event scores.

We rounded each kick's exact coordinates to the nearest centimetre. For a second set of readings, we first added independent Gaussian errors in each frame (6 mm standard deviation per coordinate) and a Gaussian bias shared by both frames (8 mm standard deviation per coordinate). Release position and frame times were exact in both sets. For noisy readings the predictor used the same error model as the generator. Appendix A treats timing errors separately.

We prespecified the three methods, coverage targets from 50 to 100% (90% for the main comparison), and subgroups by spin family and distance from the goal edge. Each kick used one inference seed across methods and measurement conditions. The predictor saw only the input positions. We saved and hashed its forecasts before scoring them against the references.

6.2 Calibrating regions on the goal grid

For each method and measurement condition, calibration sets a threshold for including cells. In the aerodynamic forecasts, rank the grid cells $j_1, \dots, j_J$ by decreasing probability $H_D(j)$ conditional on direct entry, breaking ties in a fixed order. The score of an occupied cell is the mass of the cells ahead of it,

$$S_D(j_k) = \sum_{\ell<k} H_D(j_\ell). \tag{11}$$

An empty cell scores one. For gravity, a cell whose centre lies distance d_j from the ballistic crossing scores $d_j/(L + d_j)$, where L is the goal diagonal. Thresholding this score amounts to choosing a radius about the crossing. If the map or ballistic crossing is unavailable, every cell receives a score of one.

For a target coverage c and n calibration entries, sort the scores of their true cells and take the $\lceil (n + 1)c \rceil$-th smallest as q_c. Include all cells with $S_D(j) \leq q_c$. If the rank exceeds n, set $q_c = 1$. Whenever $q_c = 1$, the region covers the whole goal. This split-calibration rule gives marginal coverage under exchangeability [8]. Our test measures the hit rate on the specified, block-balanced synthetic population. Coverage on real kicks requires independent field data.

Table 3. Forecasts at the 90% calibration target. Hits, mean area and mean point error use the 71 direct entries. The Brier score for direct entry uses all 120 test kicks. The gravity control uses both readings and a calibrated radius.

Profile	Forecast	Hits	Area (m^2)	Mean error (m)	Brier
rounded	100 ms only	64/71	2.003	0.208	0.070
rounded	50 + 100 ms	67/71	0.381	0.170	0.018
rounded	Gravity + radius	70/71	1.222	0.186	0.083
tracking	100 ms only	62/71	1.825	0.216	0.075
tracking	50 + 100 ms	60/71	0.859	0.255	0.078
tracking	Gravity + radius	69/71	1.255	0.199	0.075

6.3 The early frame narrows the region under centimetre rounding

At the 90% target, two frames covered 67/71 entries in 0.381 m^2 on average, compared with 64/71 in 2.003 m^2 using the late frame alone (Table 3). The respective 95% Wilson intervals for coverage were 86.4–97.8% and 81.0–95.1%. At their 95% targets, both methods covered 70/71 entries. The two-frame region averaged 0.585 m^2, compared with 3.482 m^2 for the late frame. Gravity covered 70/71 in 1.222 m^2 at its 90% target (Fig. 6). Equal counts can include different kicks.

Adding the early frame lowered point error on 13 entries, left it unchanged on 54 and raised it on four (Fig. 7). The mean reduction was 3.80 cm (95% interval 0.22–7.62 cm from 4,000 paired resamples of kicks). The mean reduction relative to gravity had a 95% interval of −2.29 to 5.85 cm. The Brier score for direct entry was 0.018 with both frames, 0.070 with the late frame and 0.083 for the binary gravity forecast.

6.4 Tracking error changes which forecast is more precise

At the same 90% target, the two-frame region was the smallest but covered only 60/71 noisy crossings, compared with 62/71 for the late frame and 69/71 for gravity. All eleven misses by the two-frame forecast came from the stable-spin group (12/23 hits). It covered 24/24 in each of the zero-spin and wake groups. Gravity covered 21/23 stable-spin entries (Table 6).

At 67/71 observed hits, gravity used 0.933 m^2 at its 80% target and two-frame aerodynamics used 1.385 m^2 at its 95% target. These two targets yielded equal coverage on this particular test sample. For the 71 direct entries, the aerodynamic grid centre had a mean point error 5.57 cm larger than the continuous gravity forecast (paired 95% interval 1.17–10.25 cm). The intervals for both the point-error and Brier-score differences between two-frame and late-frame aerodynamics included zero.

Estimating acceleration from the readings divides the second position difference by h^2, with $h = 50$



Figure 6. Goal area at matched test coverage. Panels A and B plot hits against mean area on a log scale for the same 71 direct entries. Open markers denote the 90% calibration target. Panels C and D compare prespecified thresholds that yielded equal hit counts. The late-frame curve has no sampled point at 67/71 hits. The full opening spans 17.86 m^2.

ms. A bias shared by the two frames survives in that difference when the release position is fixed independently (Eq. (10)). Gravity fixes the acceleration. Its relative precision here is consistent with the difficulty of estimating transverse curvature from noisy readings, even when the noise model is known.

6.5 Forecast availability and computation time

Each method produced a point for all 71 direct entries and an event probability for all 120 kicks under both measurement conditions. Only one run, a noisy late-frame forecast, fell below 25 effective samples. Even with a defined point, the noisy two-frame region filled the entire goal at its 97.5% target. Appendix B gives the counts near the goal edge.

In the local Python run, two-frame inference and goal-map construction took 84 ms at the median and 94 ms at the 95th percentile with rounding alone. With tracking errors the times were 121 and 131 ms. At the test sampling budget, 131 ms is about 44% of the nominal 299 ms remaining after the last reading. Capture, reconstruction, transfer and goalkeeper movement require additional time, and the crossing time varies by kick.

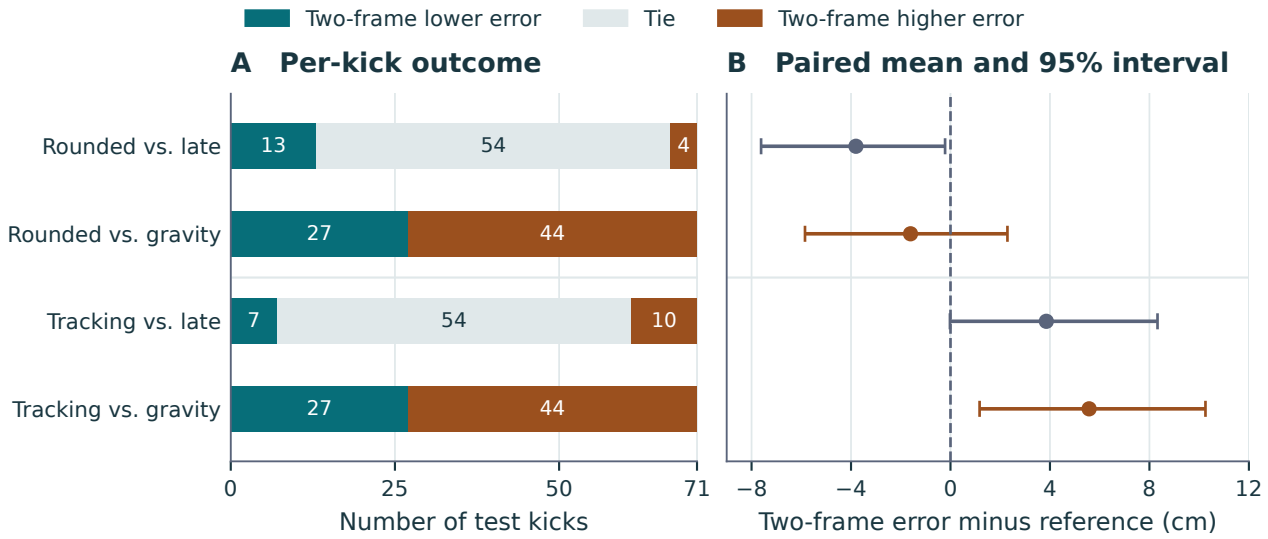


Figure 7. Point errors on the same 71 direct-entry kicks. A counts kicks on which two-frame aerodynamics has a smaller, equal or larger error than each comparison method. B shows the mean paired difference and its 95% shot-bootstrap interval. Negative values favour two-frame aerodynamics.

6.6 Stable spin in an independent test

The 12/23 aerodynamic hits on noisy stable-spin entries in Z09 prompted a second test (Z10). We drew 300 new calibration kicks and 600 new test kicks, balanced by the same three spin families, and used DOP853 to generate the reference positions and first events. Each predictor saw only the 50 and 100 ms readings. We locked the forecasts, fitted a 90% threshold for each method on the 180 calibration entries pooled across families (rank 163), then scored the held-out crossings.

With tracking errors, two-frame aerodynamics covered 343/368 direct entries overall (93.2%) and 90/110 with stable spin (81.8%). The stable-spin group accounted for 20 of the 25 misses (Fig. 8A). Gravity covered 77/110 stable-spin entries. Aerodynamics alone covered 31 and gravity alone covered 18, a paired difference of +11.8 percentage points (95% interval 0.0 to +23.6). The mean areas were 1.019 and 0.845 m^2, respectively. In a *post hoc* check, we exchanged the previously fitted Z09 and Z10 thresholds. With both forecasts fixed for each of these 110 kicks, the Z09 thresholds gave 78 aerodynamic and 94 gravity hits. Changing only to the Z10 thresholds reversed the order to 90 and 77. On the original 23 entries gravity led under either pair of thresholds (Appendix H).

Cleaner measurements improved the point forecast. On these 110 kicks, centimetre rounding alone, with a matching likelihood and calibration, reduced mean aerodynamic point error from 0.421 to 0.258 m and raised coverage to 100/110. Keeping the noisy observations and halving the prior ranges for S's transverse-force coefficient and E's spin vector gave a mean error of 0.379 m, with 94/110 hits in 0.579 m^2 after calibration. We could also expand the original noisy regions by applying the largest of the three family cutoffs to every kick. This rule needs spin labels only in calibration. It recovered 15 stable-spin entries (105/110) at a mean area of 1.843 m^2. Adjusting the cutoff changes which cells enter the region. The point prediction and event probability still come from the original forecast.

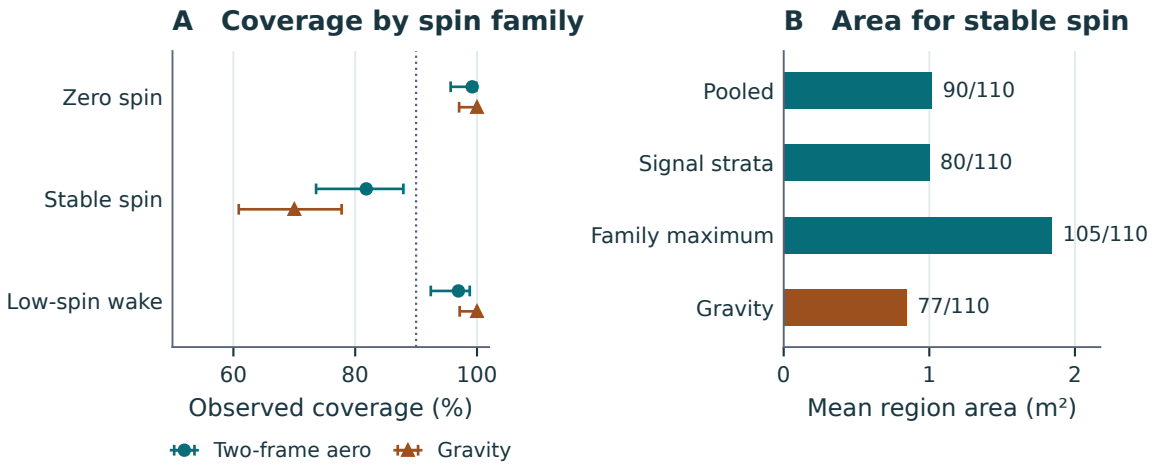


Figure 8. Two-frame regions under tracking noise in Z10. A shows coverage and 95% Wilson intervals for 127 zero-spin, 110 stable-spin and 131 low-spin wake entries. Each method uses its own pooled cutoff. The dotted line marks the 90% calibration target. B gives mean areas and hit counts on the 110 stable-spin entries. The first three bars apply a pooled cutoff, cutoffs by measured curvature, or the largest family cutoff to the same aerodynamic maps. Gravity uses its own pooled radius.

7 From a crossing forecast to a decision time

For the prescribed kick, the median arrival of 0.399 s leaves 299 ms after the last frame for both computation and movement. In a separate synthetic test, shifting the two observations from 50/100 to 100/150 ms reduced mean point error from 0.182 to 0.126 m and left 50 ms less time to act. Keeping all three frames at the later deadline gave more low-ESS forecasts (Appendix E). To translate either forecast into a save probability, one would also need the goalkeeper's initial position, movement onset and reach.

Match tracking poses a different obstacle. Three penalties recorded at 25 Hz had later crossing references, but none had an independently verified release time. The nominal force and measurement assumptions could not explain both early frames in any case. Only one late-frame estimate could be scored retrospectively (Appendix F).

The forecast describes an uninterrupted flight under specified aerodynamic priors. Ground rebounds, interventions and collisions with the posts or crossbar could change the outcome near the goal edges, and the mix of forces in actual matches may differ from the synthetic population. The grid, measurement model and decision time therefore belong alongside any quoted point and region.

Conclusion

For the prescribed kick, the nominal modal cell lies at $(Y, Z) \approx (2.56, 2.07)$ m. Conditional on direct entry, six cells spanning 2.196–2.928 m laterally and 1.708–2.196 m vertically contain 90.52% of the model probability over 0.3572 m^2. Even flights that agree with both early readings can cross 8.76 cm apart.

With centimetre rounding, adding the 50 ms reading reduced the area needed to cover 70 of 71 entries. Under tracking noise, gravity used less area at 67 of 71. In the larger independent test, stable-spin coverage fell below the pooled 90% target. Cleaner measurements improved point accuracy. A higher cutoff recovered missed entries by expanding the region. Match validation requires independently timed releases, calibrated early positions and reference crossings.

Appendix A How timing and coordinate errors move the crossing

A.1 From a position error to a goal-plane error

A position error can change both where and when the ball crosses the goal plane. We can separate these effects in the constant-acceleration fit. For a fixed release position and readings at h and $2h$, its acceleration and launch velocity are

$$\widehat{\boldsymbol{a}} = \frac{\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0}{h^2}, \qquad \widehat{\boldsymbol{v}}_0 = \frac{\boldsymbol{r}_1 - \boldsymbol{r}_0}{h} - \frac{h}{2}\widehat{\boldsymbol{a}}. \tag{12}$$

The prescribed readings give $\boldsymbol{v}_0 = (30.4, 9.4, 6.9)$ m/s and $\boldsymbol{a} = (-16, -16, -12)$ m/s^2. Extrapolate with this acceleration and write $X = 11 - x_0$, $D = v_{0x}^2 + 2a_xX$. For a first forward crossing with $D > 0$, the positive root has the numerically stable form

$$T = \frac{2X}{v_{0x} + \sqrt{D}}, \quad \dot{x}(T) = \sqrt{D}. \tag{13}$$

The control crosses at $T = 0.4050$ s and $(Y, Z) = (2.495, 1.920)$ m. The aerodynamic forecast appears in Eq. (6).

Let $\delta\boldsymbol{r}(T)$ be the change in position at the original crossing time. Requiring the perturbed trajectory to satisfy $x = 11$ m at its own crossing gives

$$\delta T = -\frac{\delta x(T)}{\dot{x}(T)}, \qquad \delta Q = P\,\delta\boldsymbol{r}(T), \qquad P = \begin{pmatrix} -\dot{y}(T)/\dot{x}(T) & 1 & 0 \\ -\dot{z}(T)/\dot{x}(T) & 0 & 1 \end{pmatrix}. \tag{14}$$

The local slopes convert longitudinal error into a shift across the goal. An error parallel to the velocity changes only the arrival time to first order, since $P\boldsymbol{v}(T) = 0$. The slopes can diverge as $\dot{x}(T)$ approaches zero. If the forward crossing disappears, its location derivative is undefined.

The Lagrange weights of the two readings are $\lambda_1 = s(2 - s)$ and $\lambda_2 = s(s - 1)/2$, where $s = T/h$. Let $\boldsymbol{e}_i$ be the spatial error in reading i. A common time offset $\tau > 0$ means that we fit the readings at their labelled times $h, 2h$ although both were taken τ later. To first order, the combined shift is

$$\delta Q = P\left[\lambda_1\boldsymbol{e}_1 + \lambda_2\boldsymbol{e}_2 + \big(\lambda_1\boldsymbol{v}(h) + \lambda_2\boldsymbol{v}(2h)\big)\tau\right]. \tag{15}$$

At the nominal crossing the weights are -49.41 and 28.76. Allowing independent errors of up to 5 mm in each coordinate gives linear bounds of 0.439 m in Y and 0.424 m in Z. A common $+1$ ms offset shifts (Y, Z) by $(-0.123, -0.093)$ m to first order. Central differences with a $1\,\mu$s step reproduce the timing derivative to within 2×10^{-5} m/s. The large weights arise from extending an unconstrained quadratic well beyond the observations. The aerodynamic predictor restricts curvature through its force and launch bounds.

A.2 Covariance near a smooth crossing

Let Σ_e be the covariance of noise drawn independently in each frame, Σ_b that of a bias shared by the frames, and σ_τ^2 the timing variance. Assume independence among these sources. With $w = \lambda_1\boldsymbol{v}(h) + \lambda_2\boldsymbol{v}(2h)$, Eq. (15) gives

$$\mathrm{Cov}(\delta Q) = P\left[(\lambda_1^2 + \lambda_2^2)\Sigma_e + (\lambda_1 + \lambda_2)^2\Sigma_b + \sigma_\tau^2 w w^\top\right]P^\top. \tag{16}$$

Timing uncertainty acts along a single direction, Pw. The bias coefficient changes if the release shares the frames' translation. This covariance describes measurement uncertainty near a smooth crossing. Model discrepancy and sampling error contribute separately.

A.3 Joint perturbations and loss of a forward crossing

We combined both timing limits and zero offset with every corner of the six-coordinate error box, counting a zero bound once. For each combination we refitted the quadratic and solved for its first forward plane crossing, ignoring the ground and goal edges. The largest displacement among sampled crossings need not bound the full box, whose interior may contain larger shifts.

Table 4. Joint perturbations of the quadratic control. The two linear bounds sum the absolute derivative contributions in each coordinate. The sampled shift is the largest Euclidean displacement among forward crossings. The last column counts trials with no forward root among all tested combinations.

Phase bound (ms)	Spatial bound (mm)	First-order Y (m)	First-order Z (m)	Sampled shift (m)	No root (count)
0	0	0.000	0.000	0.000	0/1
0	2.5	0.219	0.212	0.319	0/64
0	5	0.439	0.424	0.671	0/64
0	10	0.877	0.848	1.502	0/64
1	0	0.123	0.093	0.197	0/3
1	2.5	0.342	0.305	0.533	0/192
1	5	0.562	0.517	0.930	0/192
1	10	1.000	0.942	2.025	0/192
3	0	0.370	0.280	0.297	1/3
3	2.5	0.589	0.492	0.537	64/192
3	5	0.808	0.704	2.298	48/192
3	10	1.247	1.128	4.487	32/192

With bounds of 1 ms and 5 mm, all 192 combinations crossed the plane. The largest sampled displacement was 0.930 m, while the separate linear bounds in Y and Z were 0.562 and 0.517 m (Table 4). At 3 ms and 5 mm, 48/192 combinations lost the forward root. Widening a box changes the corners sampled, so the failure fraction can rise or fall. Propagating each perturbation to its own crossing captures both large shifts and the loss of the event.

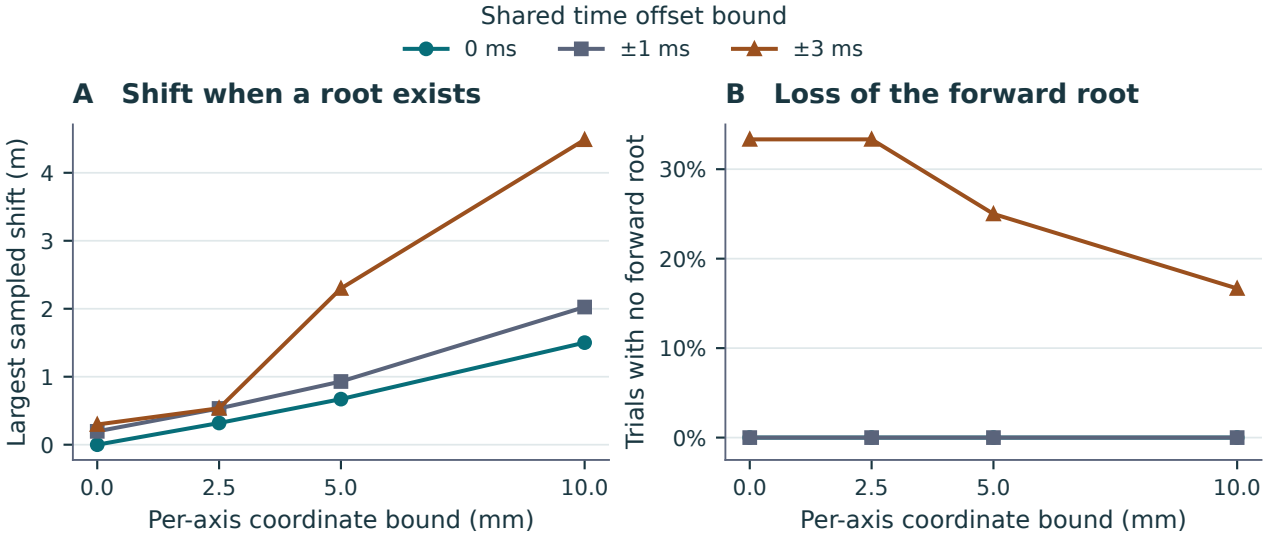


Figure 9. A perturbation can shift or remove the plane crossing. A gives the largest displacement among sampled quadratic fits with a forward root. B gives the fraction of tested combinations with no root. Each timing and coordinate combination is a deterministic trial.

Appendix B Coverage by subgroup and computation time

Mean region area includes all 71 direct entries among the 120 test kicks. A calibration target of 100% returns the full opening by construction.

Table 5. Observed hits and mean area at selected calibration targets. Both measurement conditions use the same 71 direct entries.

Profile	Forecast	Target (%)	Hits	Mean area (m^2)
rounded	100 ms only	80	53/71	1.367
rounded	100 ms only	90	64/71	2.003
rounded	100 ms only	95	70/71	3.482
rounded	100 ms only	100	71/71	17.861
rounded	50 + 100 ms	80	58/71	0.187
rounded	50 + 100 ms	90	67/71	0.381
rounded	50 + 100 ms	95	70/71	0.585
rounded	50 + 100 ms	100	71/71	17.861
rounded	Gravity + radius	80	68/71	0.917
rounded	Gravity + radius	90	70/71	1.222
rounded	Gravity + radius	95	71/71	2.296
rounded	Gravity + radius	100	71/71	17.861
tracking	100 ms only	80	59/71	1.203
tracking	100 ms only	90	62/71	1.825
tracking	100 ms only	95	69/71	2.569
tracking	100 ms only	100	71/71	17.861
tracking	50 + 100 ms	80	54/71	0.523
tracking	50 + 100 ms	90	60/71	0.859
tracking	50 + 100 ms	95	67/71	1.385
tracking	50 + 100 ms	100	71/71	17.861
tracking	Gravity + radius	80	67/71	0.933
tracking	Gravity + radius	90	69/71	1.255
tracking	Gravity + radius	95	71/71	2.527
tracking	Gravity + radius	100	71/71	17.861

Table 6. Hits at the 90% target, by generator family and distance to the edge of the opening. Each grouping partitions the 71 entries. All subgroups use the threshold fitted for that method and measurement condition.

Profile	Stratum	Entries	100 ms	50 + 100 ms	Gravity
rounded	no spin	24	24/24	24/24	24/24
rounded	stable spin	23	16/23	21/23	22/23
rounded	wake	24	24/24	22/24	24/24
rounded	near edge	11	9/11	10/11	11/11
rounded	interior	60	55/60	57/60	59/60
tracking	no spin	24	24/24	24/24	24/24
tracking	stable spin	23	14/23	12/23	21/23
tracking	wake	24	24/24	24/24	24/24
tracking	near edge	11	9/11	7/11	10/11
tracking	interior	60	53/60	53/60	59/60

We classify a crossing as near-edge if its distance to the nearest of the four goal edges is below one nominal cell width (0.244 m). The generator family labels are used only to score subgroups. Each

kick contributed one inference seed per measurement condition. We matched differences by kick and resampled 4,000 times, using all kicks for event scores and direct entries for point error, area and hits. The intervals describe the sampled mixture of kick families, balanced before outcomes were known. Field calibration would require independently timed releases, reference crossings and a protocol for intervening contacts.

Table 7. Availability and elapsed time in one local run. Aerodynamic timings cover inference and preparation of the goal-cell masses for ranking. Gravity timings cover the ballistic solution. Both exclude application of the calibrated threshold.

Profile	Forecast	Points	Events	ESS < 25	Median (ms)	95th (ms)
rounded	100 ms only	71/71	120/120	0	90.9	103.1
rounded	50 + 100 ms	71/71	120/120	0	83.6	93.7
rounded	Gravity + radius	71/71	120/120	0	0.2	0.3
tracking	100 ms only	71/71	120/120	1	107.5	117.4
tracking	50 + 100 ms	71/71	120/120	0	121.4	130.7
tracking	Gravity + radius	71/71	120/120	0	0.2	0.3

Appendix C Forecast algorithm and physical checks

C.1 From observations to first-event probabilities

The inputs are the release point, observed positions and times, the camera-error model and the parameter ranges. Each observation must have a valid time index and finite coordinates.

1. Draw wind and aerodynamic parameters for each force law. Propose launch velocities by inverting the latest observed position, mixing in 2% draws from the launch prior to retain support elsewhere. With only the release point, sample directly from that prior.

2. Propagate each trajectory through the available observations. A trajectory that reaches the goal plane or touches down before the last reading receives zero weight. Use the observation likelihood at all supplied times in Eq. (5). If release timing is uncertain, divide the fixed budget among quadrature nodes and combine posterior masses with their quadrature weights.

3. Propagate the surviving trajectories to the first forward crossing, descending ground contact or time horizon. Normalize each event mass by the total weight. A crossing in the open goal rectangle contributes to its goal cell. Divide each cell mass by the total direct-entry weight.

4. Return the event probabilities, modal cell, raw region and effective sample size. Zero total weight leaves the event distribution undefined. If only the direct-entry weight is zero, the event probabilities remain available and the conditional point and map are undefined.

Report the unconditional entry probability alongside the conditional point, especially near the goal boundary where entry mass may be small. Geometric bounds can exclude a flight under the stated assumptions. A low effective sample size flags uncertainty in the numerical approximation.

C.2 Assumptions, limits and solver checks

Under gravity alone, $x(t) = v_{0x}t$, $y(t) = v_{0y}t$ and $z(t) = R + v_{0z}t - gt^2/2$. The analytic plane and ground times check event ordering. In still air, drag alone preserves horizontal heading because the

Table 8. Inputs used in the nominal forecast and in the tracking-error test. A change to an assumed range or error model requires a new forecast.

Quantity	Role	Interpretation
Release, frame times and opening	Given by the problem	Fix the observations and geometry
Ball mass, radius and air density	Nominal physical inputs	Set the factor $\rho\pi R^2/(2m)$
Empirical drag curve	Literature-based fit	Limited speed range, with clipping and an offset prior
Wind and lift or spin bounds	Scenario choices	Restrict admissible flights and affect model weights
5 mm coordinate half-width	Rounding assumption	Links continuous positions to displayed readings
Cell width and RK4 steps	Numerical choices	Affect the modal cell and integration accuracy
6 mm noise and 8 mm bias	Synthetic test settings	Specify independent and shared tracking errors

horizontal acceleration is proportional to the horizontal velocity. Equation (2) extends this property to the frame of a constant wind. These limits check the geometry and force signs independently of posterior sampling.

Four RK4 comparisons with adaptive DOP853 cover vacuum flight, drag with ground contact, effective transverse lift and the empirical force law. The solvers agreed to within 0.171 mm in event position and 0.038 ms in event time on these cases. The trajectory pair in Sec. 4 also has a separate high-accuracy integration and MATLAB replay. The observations still limit what can be inferred about wind, spin and release time.

Appendix D Reference flights and earlier synthetic tests

D.1 The reference generator and kinematic controls

To test how the forecast responds to imperfect force laws, we generated reference flights with a different drag curve,

$$C_D^{\mathrm{ref}} = 0.19 + \frac{0.20}{1+\exp[(U-20)/2.5]} + d_0, \qquad d_0 \sim \mathcal{U}(0, 0.1),$$

and a lift coefficient $0.32\tanh(4R\|\boldsymbol{\omega}\times\boldsymbol{u}\|/U^2)e^{-0.12t}$ acting along $\boldsymbol{\omega}\times\boldsymbol{u}$. Launch-velocity components were independently uniform on $[20, 36]$, $[-11, 11]$, $[2, 10]$ m/s. Wind was uniform on a horizontal disk of radius 2.5 m/s. We used equal numbers of kicks with zero spin, stable spin and a low-spin wake. Their spin-component bounds were 0, 55 and 6 rad/s. Wake flights also had a time-varying acceleration given by projecting $A(0, \sin\phi(t), 0.4\cos\phi(t))$ perpendicular to $\boldsymbol{u}$. We chose $A \in [2, 5]$ m/s^2, a period of 0.18–0.55 s and a uniform initial phase. Experiments motivate including unsteady low-spin forces [6]. The amplitudes, periods and family proportions here are synthetic.

The original study used DOP853 [4] and split whole kicks into 9 for development, 36 for calibration and 48 for blind testing. Follow-up tests through Z09 used new seeds with the same reference force laws and a 1 ms RK4 step. Z10 used DOP853 for its new reference flights. On the original 48 DOP853 flights, RK4 reproduced every first event, with maximum differences of 1.9×10^{-6} m at the crossing and 6.3×10^{-7} s in event time. We gave each predictor only its designated early frames and saved and hashed its forecasts before scoring. These comparisons test numerical agreement and performance on the simulated kick population.

On true direct entries we report Euclidean point error, region hits and mean region area. We score the probability of direct entry across first events with the Brier score $(P(G) - \mathbf{1}_G)^2$. We compare against a least-squares fit under gravity alone and a constant-acceleration fit through the three positions. We count missing event forecasts separately from missing conditional points.

D.2 The early frame at a fixed sampling budget

Of the original 48 blind-test kicks, 33 entered directly, ten crossed outside the opening and five hit the ground first. The two-frame method had median point error 0.133 m and raw-region coverage 30/33, using eight times the single-frame proposal budget. We revisited the same kicks with 2,048 proposals per model for each input choice and three paired seeds (Table 9).

Table 9. Retrospective test at 100 ms with equal proposal budgets. The 33 direct entries and three seeds give 99 point forecasts. Both methods returned event estimates on all 144 attempts. Regions use raw probabilities.

Observed frames	Mean error (m)	Hits	Area (m^2)	ESS < 25
100 ms only	0.215	99/99	3.071	0
50 and 100 ms	0.170	91/99	0.288	11

Adding the 50 ms reading narrowed the regions and reduced their coverage. We averaged paired differences over seeds within each kick, then resampled the 33 kicks 2,000 times. The mean point-error difference (both frames minus the late frame) was -4.46 cm, with a descriptive 95% bootstrap interval of $[-9.14, 0.08]$ cm. The equal-budget result supports a reduction in area. The interval still admits zero improvement in mean point error.

D.3 When tracking noise exceeds the rounding interval

We applied three likelihoods to identical noisy positions on a separate set of 36 calibration and 48 test kicks. Before centimetre rounding we added an 8 mm Gaussian bias shared across frames for each axis and independent 6 mm Gaussian frame errors. All aerodynamic forecasts used 8,192 proposals per model and two paired seeds. One likelihood treated the combined 10 mm marginal error as independent, one used the simulated 6/8 mm decomposition, and one allowed only rounding. The two tracking models used a local velocity-proposal half-width of $0.026\,\mathrm{m}/t$. The rounding-only model used $0.007\,\mathrm{m}/t$. Importance weights corrected each proposal. The finite-budget comparison therefore includes both the likelihood and the proposal width.

Table 10. Noisy-input tests at 100 ms on 36 direct entries and two seeds. n_Q counts available point forecasts. Region hits and mean area use all 72 attempts. An undefined map gives an empty raw region. A calibrated region can encompass the whole opening.

Likelihood	Mean error (m)	n_Q	Raw hits	Raw area (m^2)	Calibrated hits	Cal. area (m^2)
Rounding only	0.425	68	27/72	0.355	72/72	17.861
Independent error	0.231	72	72/72	1.870	68/72	1.261
Shared + independent	0.257	72	71/72	1.256	71/72	1.609

We calibrated each method and seed separately using the prefix-mass score in Sec. 6. A true cell with zero mass receives score one. With n calibration entries, we include every cell whose score is at most the $\lceil 0.9(n+1) \rceil$th ordered calibration score. A cutoff of one or a rank above n returns the whole opening. Coverage is empirical here because the design balances synthetic families and the

field error distribution is unknown.

The rounding-only method covered all 72/72 entries after calibration by returning the entire 17.861 m^2 goal. The shared-bias likelihood reduced mean point error relative to rounding alone by 15.86 cm on 34 complete paired kicks (95% shot-bootstrap interval 4.95–26.53 cm). Its interval against the independent-error model includes zero. Gravity had lower mean point error (0.220 m) than either tracking model, but its event Brier score was higher (0.104 versus 0.069 for independent errors and 0.064 for shared bias). The rounding-only Brier score was 0.101 on 94/96 attempts. The other methods supplied all 96 event estimates.

Appendix E The value and cost of waiting for another frame

E.1 Moving the last frame or keeping an extra one

Using these same calibration and test kicks with rounding alone, we compared three sets of observations at 8,192 proposals per model and two paired seeds. Moving from the 50/100 ms pair to the 100/150 ms pair delays the decision. Comparing that later pair with all three frames holds the 150 ms deadline fixed.

Table 11. Additional observations on 36 direct entries with two seeds (72 attempts). n_Q counts available point forecasts. An undefined map contributes a miss and zero area. Missing event estimates and low ESS use all 96 attempts.

Frames (ms)	Mean error (m)	n_Q	Hits	Area (m^2)	No event estimate	ESS < 25
50, 100	0.182	72	61/72	0.346	0	0
100, 150	0.126	72	62/72	0.136	0	20
50, 100, 150	0.130	65	52/72	0.093	6	46

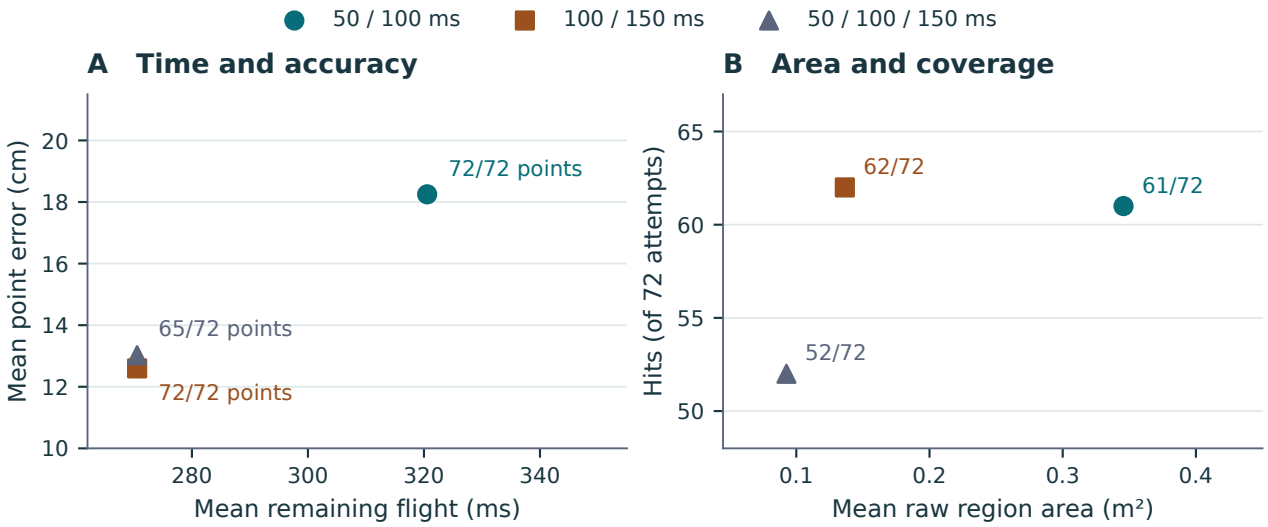


Figure 10. Forecast accuracy and time left to act. A shows mean point error for available direct-entry estimates (65/72 for three frames) and mean remaining flight time over all 36 direct entries, before computation or goalkeeper movement. B shows raw region area and coverage, with undefined maps counted as empty (Table 11).

Waiting for the later pair reduced mean point error by 5.65 cm on 36 complete paired kicks (95%

shot-bootstrap interval 2.79–8.98 cm). Mean time remaining fell from 321 to 271 ms. Calibrated coverage rose from 61/72 to 64/72 while mean area fell from 0.350 to 0.177 m^2.

At the same 150 ms deadline, keeping all three frames left six of 96 attempts without an event estimate and 46/96 with low ESS. Both seeds yielded paired points on only 31 kicks. On those kicks the three-frame mean point error exceeded the later pair's by 0.27 cm (95% interval $[-1.71, 2.28]$ cm). The three-frame raw regions were smaller but covered only 52/72 entries. Few proposals fit all three readings. This limits what the comparison can establish about the earliest frame's information content. Model error and finite-test variation may also contribute. An adaptive or sequential sampler would need its own validation.

E.2 When a later frame distinguishes the two fitted flights

At 150, 200 and 250 ms, the largest coordinate gaps in the pair from Sec. 4 are 1.35, 5.30 and 12.96 mm (Fig. 4B). Deterministic centimetre rounding already gives different heights at 200 ms, 1.26 and 1.25 m. If each coordinate can carry any error within ±5 mm, the two reading boxes still overlap at that time. Of the three times tested, only 250 ms distinguishes the flights for every allowed error.

E.3 Which frames reach the predictor in time?

We simulated 13 camera and noise conditions on a separate set of 48 kicks, 16 per force family and 33 direct entries. At each deadline on the estimated release clock (50 and 100 ms), we selected the latest distinct frame already received and at most 10 ms old. The predictor used its native timestamp. With 2 ms delivery delay, rates of 25, 60, 120 and 240 Hz supplied both readings on 0, 22, 47 and 48 kicks, respectively. At 240 Hz, increasing the delay to 6 ms reduced availability to 46/48. These counts depend on the simulated phase and frame-age rule. Computation takes additional time.

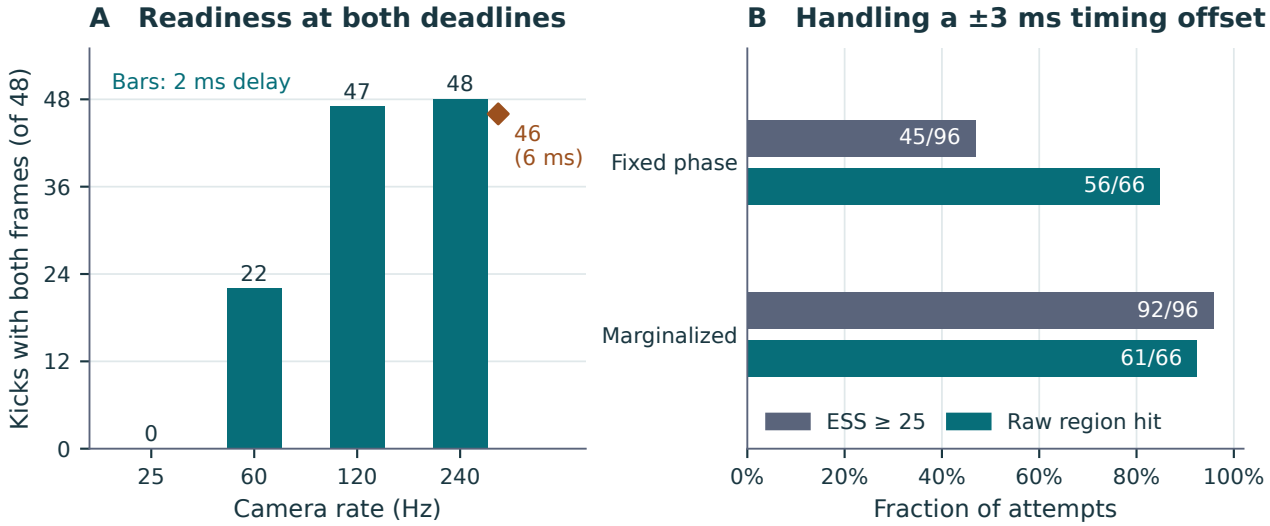


Figure 11. Timely frames and uncertain release times. A counts how many of the 48 kicks supplied both frames on time. B compares a fixed release estimate with integration over the simulated ±3 ms offset. ESS uses 96 attempts (48 kicks and two seeds). Raw-region coverage uses 66 direct-entry attempts.

For periodic frames delivered with fixed delay ℓ and no drops, the oldest possible latest frame has age bounded by $\ell + 1/f$, where f is the frame rate. To guarantee the 10 ms age limit at 2 ms delivery delay requires at least 125 Hz. A 240 Hz camera allows more margin, but consecutive exposures alone bracket release only to a half-width of about 2.08 ms. A 1 ms timing target would require an independent, finer time reference.

Integrating over the same simulated ±3 ms release offset increased the number of forecasts with ESS at least 25 from 45/96 to 92/96. Raw regions covered 61/66 entries, up from 56/66. They also grew from 0.377 to 0.543 m^2 on average, and mean point error rose from 0.196 to 0.218 m. These simulated conditions provide targets for a field trial.

Appendix F Compatibility with 25 Hz match tracking

F.1 Reference crossings on the match clock

The seven matches in the IDSSE v1 elite-soccer dataset contained three executed penalties, all recorded as goals [9]. We translated the longitudinal coordinate by 41.50 m into the penalty-spot frame. We treated the first live frame after a dead-ball interval as an exact anchor and evaluated forecasts at +80 ms using the original priors and a 5 mm coordinate bound. The inputs were either the +80 ms frame alone or the pair at +40 and +80 ms. Reference crossings were interpolated from later frames in the same system, at least 0.32 s after the cutoff. We locked the forecasts before scoring, but had seen the crossings while screening the data.

Table 12. Three penalties on the match clock. The tracked crossing is the reference. The displacement or second difference in the third column yields the required minimum acceleration under the launch assumptions. Here $\Delta^2 x = x_2 - 2x_1 + x_0$.

Fixture	Reference (Y, Z) (m)	Observed quantity (m)	Min. accel. (m/s^2)
Cologne–Bayern	$(2.750, 0.160)$	$z_1 - z_0 = -0.030$	106.25
Düsseldorf–Regensburg	$(2.919, 1.179)$	$\Delta^2 x = 0.510$	309.38
Düsseldorf–Kaiserslautern	$(3.070, 0.466)$	$\Delta^2 x = 1.100$	678.13

F.2 An acceleration bound for consecutive positions

In constant wind the energy balance gives $\dot{U} \le g$, so $U \le \sqrt{38^2 + 12^2 + 11^2} + 3 + 9.81(0.080) = 45.125$ m/s over the first 80 ms. Drag and lift are orthogonal in Eq. (3). Combining them in quadrature and adding g gives upper acceleration bounds of 58.681, 89.170 and 102.648 m/s^2 for D, S and E, using $C_D \le 0.771$ and $C_L \le 0.38$ in E. Thus $A_* = 102.648$ m/s^2 bounds all three models.

For three coordinates q_0, q_1, q_2 at spacing $h = 0.040$ s, the true second difference integrates acceleration against a triangular kernel of area h^2. An exact anchor and errors bounded by ϵ in the two later positions therefore require

$$|q_2 - 2q_1 + q_0| \le A_* h^2 + 3\epsilon, \qquad z_1 - z_0 \ge v_{z,\min} h - \tfrac{1}{2} A_* h^2 - \epsilon. \tag{17}$$

The second inequality uses the launch prior $v_{z,\min} = 1.5$ m/s. Regensburg's successive x-increments of 0.71 and 1.22 m require at least $(0.510 - 0.015)/h^2 = 309.38$ m/s^2. Kaiserslautern's record repeats the anchor and then advances 1.10 m. Cologne misses the vertical bound by just 2.9 mm. All three exceed the common acceleration cap under the assumed launch conditions (Table 12).

Treating the anchor as a ball already in flight helps distinguish these failures. Relaxing the lower bound on vertical velocity from +1.5 to −11 m/s gives Cologne two forecasts with ESS 58 and 48. The other two records remain unsupported. Cologne is sensitive to the assumed initial velocity, and its release time remains unknown.

F.3 A forecast from the late frame alone

Using only the +80 ms position, neither seed gave positive weight for Cologne or Kaiserslautern. Both placed the Regensburg modal cell at (2.074, 1.098) m, 0.849 m from the reference. The first raw region covered 4.941 m^2 and contained the reference. The second covered 4.584 m^2. With the same late frame, gravity and straight-line fits missed by 0.809 and 1.219 m. With both frames, gravity missed by 0.853 m at Regensburg and predicted ground contact first in the other two cases.

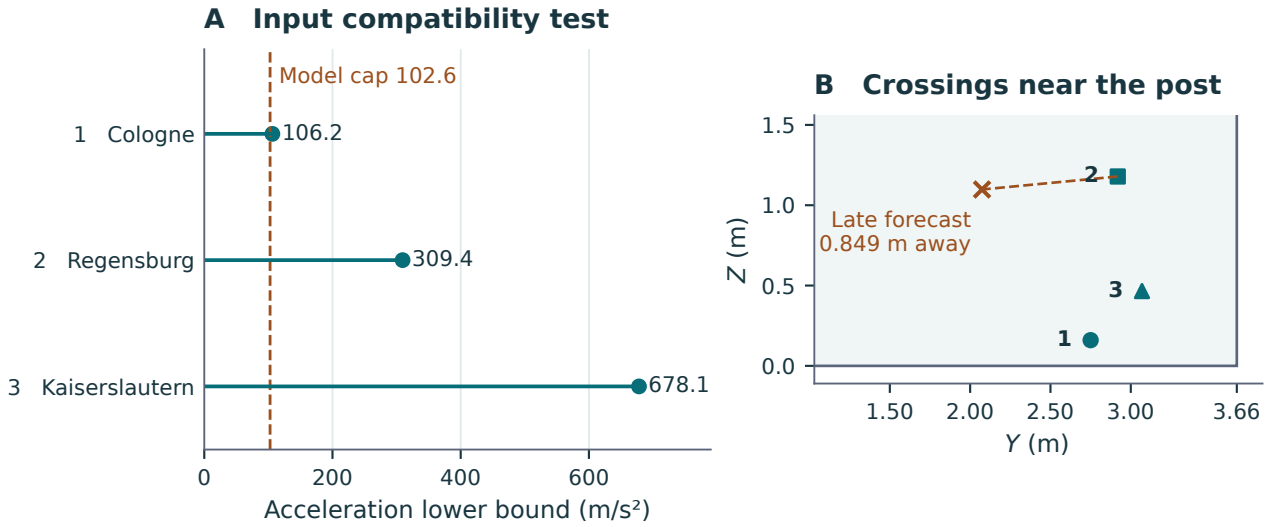


Figure 12. Testing the launch assumptions against three match records. A compares the required minimum accelerations with the common model cap. B locates the three later crossing references near the positive-Y post. The cross marks the only available late-frame aerodynamic forecast. The records leave release timing and the first event unverified.

Appendix G Evidence needed for a field test

G.1 What the available tracking data resolve

Regensburg's anchor lay about 7 cm horizontally from the penalty spot. The other two anchors were already 0.71 and 1.47 m downfield. No kick had a release time independently verified within 10 ms or a direct entry confirmed as its first event. The acceleration test identifies incompatible inputs and assumptions. It cannot distinguish a mid-flight starting frame from a timing or tracking error. The single scoreable late-frame map gives no estimate of field coverage.

ISSIA-3D provides a separate multi-camera reconstruction of one match sequence [10]. Of 79 continuous segments, 61 supplied two positions 80 ms apart and 47 retained the ball for another 120 ms. On those 47, constant-velocity extrapolation gave a median three-dimensional error of 0.162 m. Holding the last position gave 0.996 m. These were subsequent positions in ordinary play. Penalty testing also needs an identified release and a first event at or before the goal plane.

G.2 A release-synchronized validation protocol

Independently annotate contact in synchronized multi-view footage and record clock alignment, coordinate calibration, lost frames and intervening contacts. For a scoreable kick, both early positions must be available by the cutoff and the first event must be identifiable. Record interceptions before the natural crossing as contacts or censored outcomes.

Use independent calibration sequences to estimate timing bias, per-axis error and cross-frame

dependence in Eq. (8). The three existing cases can guide development. Freeze the model before testing on held-out matches. Report forecast availability among eligible kicks, point error, joint coverage and area on the same direct entries, and Brier scores on confirmed first events.

A marked-ball field test can record spin and deflection together using multi-view tracking of surface markers [11]. Independently time release and the crossing on the same kick to test coverage by spin.

The nominal 0.399 s schedule leaves 299 ms after the 100 ms reading and only 149 ms after a 250 ms reading. A field test of additional information such as spin or wind should use the same kicks and count the time needed to obtain each measurement.

G.3 Scope of the evidence

The wind inequality and the fitted pair establish limits under specified force and measurement assumptions. Solver comparisons test the integration of those equations. The synthetic holdouts measure how well forecasts transfer to different prescribed force laws. The match records expose conflicts with launch and timing assumptions. Measuring field accuracy still requires independent release and crossing references.

The ball-centre model omits deformation, finite-radius collisions with posts or crossbar, rebounds and goalkeeper contacts. Model weights depend on unmeasured priors and the empirical coefficients need not match the ball. Only the reference generator includes a low-spin wake. One-factor tests leave joint extremes unresolved. Repeated seeds change sampling variability without adding independent kicks. The sampler has been tested offline, and camera readiness alone does not measure end-to-end delivery time.

A forecast should identify its decision time and input frames, report first-event probabilities, and state the point convention and joint region whenever direct-entry mass is positive. Coverage, area, sampling support and the frequency of unavailable forecasts show how much information those outputs carry.

Appendix H Why two-frame forecasts miss some stable-spin entries

In Z10, both the nominal aerodynamic forecast and gravity gave points and regions for all 368 direct entries. With tracking errors, their Brier scores for direct entry were 0.064 and 0.093 over all 600 test kicks. Among the 110 stable-spin entries, the 95% Wilson intervals for coverage were 73.6–87.9% for aerodynamics and 60.9–77.8% for gravity. We calculated paired 95% intervals by resampling whole kicks 4,000 times with the calibration set fixed. The generator balanced the families before flight. Restricting comparisons to direct entries can leave different launch conditions in each family.

H.1 What the two readings tell us about acceleration

For an exact release point and equally spaced observations, write $\widehat{\boldsymbol{a}} = (\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0)/h^2$, with $h = 0.050$ s. Approximating rounding to step q by independent uniform errors gives the per-axis variance

$$\sigma_a^2 = \frac{5(\sigma_e^2 + q^2/12) + \sigma_b^2}{h^4}.$$

The independent errors have squared coefficients $1 + 4 = 5$, and the shared frame bias has coefficient $1 - 2 = -1$. The resulting standard deviation is 2.58 m/s^2 for centimetre rounding alone and 6.76 m/s^2 with the added tracking errors. Across all 200 stable-spin kicks, the observed root-mean-square errors in the second difference, pooled over the y, z components, were 2.56 and 6.47 m/s^2. The

aerodynamic point forecast on the 110 direct entries improved by 16.3 cm on average (95% interval 12.7–20.0 cm) when using rounding alone, with a likelihood and proposal width matched to each observation condition.

With release and the noisy 100 ms position fixed, a change $\delta\boldsymbol{a}$ in the second difference changes a constant-acceleration trajectory at a fixed time t by $t(t-2h)\,\delta\boldsymbol{a}/2$. At 0.40 s, 1 m/s^2 corresponds to 6 cm. We substituted the exact simulated second differences on *all three axes* and solved again for the first plane crossing or ground contact. The forward acceleration can change the crossing time. Of the 110 true entries, 104 had plane crossings in both fits. For these the mean crossing error fell by 45.2 cm (95% interval 38.4–52.8 cm). Five had a crossing only after substitution and one only before. This sensitivity calculation requires exact positions available to the simulator.

H.2 Changing the prior and fitting its threshold

We multiplied the component half-widths of both $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E by 0.5 or 1.5. A separate test changed only the D/S/E prior weights from 0.20/0.40/0.40 to 0.02/0.49/0.49. Table 13 shows which effects survive when every prior uses the same threshold. Halving both ranges improved the true cell's mean rank from 11.92 to 6.31 and reduced mean point error by 4.17 cm (95% interval 2.15–6.28 cm). After refitting the cutoff, the corresponding hit-rate difference was +3.6 percentage points (95% interval −3.6 to +10.9). The joint change leaves the individual contributions of S and E unresolved.

Table 13. Aerodynamic priors on the same 110 noisy stable-spin entries. The two hit columns use each prior's own pooled 90% cutoff and the nominal prior's cutoff (0.8594). Areas use each prior's own cutoff. Rank one denotes the most probable goal cell.

	Hits out of 110		Mean area	Mean error	Mean rank
Prior	Own cutoff	Nominal cutoff	(m^2)	(m)	of true cell
Nominal	90	90	1.019	0.421	11.92
Half-widths ×0.5	94	99	0.579	0.379	6.31
Half-widths ×1.5	82	68	1.315	0.452	15.25
Higher S/E weights	101	94	1.592	0.405	11.65

H.3 Applying alternative cutoffs to a fixed forecast

The stable-spin calibration entries gave the largest family cutoff (0.9698, rank 51 of 55). A rule that knows each test kick's true family could use its corresponding cutoff. The maximum rule in Fig. 8B applies 0.9698 to every kick. These rules agree on stable-spin entries. Across all 368 entries, using the maximum raised mean area from 1.003 to 1.839 m^2. We also assigned each kick to one of two calibration groups by comparing the measured signal $\|(\widehat{a}_y, \widehat{a}_z + g)\|/\sigma_a$ with a prespecified boundary of 2. This rule uses only the two readings. Relative to pooling, it gained seven stable-spin hits and lost seventeen, covering 80/110 in 1.001 m^2 on average.

In the post hoc threshold exchange, the 55 pooled Z09 calibration entries gave cutoffs of 0.8214 for aerodynamic prefix mass and 0.0800 for the gravity distance score. The 180 pooled Z10 entries gave 0.8594 and 0.0687, respectively. The aerodynamic cutoff rose, admitting more cells, and the gravity radius shrank. On the original 23 stable-spin entries the counts changed from 12 and 21 to 14 and 19. On the 110 newer entries the threshold source alone determined which model covered more kicks (Sec. 6.6).

Appendix I Reproducing the forecasts

The source archive records the prescribed-case inputs, seeds 8111/8117, the 30×10 grid and solver tolerances. The equal-budget replay uses three seeds. The additional-frame and noise tests use truth seeds 77401/77407 and inference seeds 77501/77507. Tables and figures draw on frozen outputs, with checksums for the benchmark source and predictions. Reproducing the prescribed forecast requires no match data.

Claim or check	Package evidence
Wind geometry and model assumptions	`design/`, `report/methods.md`
Feasible pair and prescribed prediction	`experiment/runs/B01/`, `B03/`
Equal-budget comparisons	`experiment/runs/Z04/`, `report/evidence/z09-replay.zip`
Stable-spin test and calibration	`report/evidence/z10-replay.zip`
Conditional factor sensitivity	`report/evidence/z05-factors.csv`
Timing and noisy-input tests	`report/evidence/z06-summary.json`
Match timing and anchor checks	`report/field.md`, `experiment/runs/Z07/`
Causal acquisition simulations	`report/evidence/z08-feedback.csv`
Generated tables	`paper/build_assets.py`, `build_revision_assets.py`
Vector figures and plotted data	`paper/build_visuals.py`, `visual-data.json`

From the repository root, run `python report/recheck.py`, `python report/recheck_field.py` and `python report/recheck_extensions.py` for the headline checks. The two table builders and `python paper/build_visuals.py` regenerate the assets used by `paper/manuscript.tex`. The figure builder verifies the goal-cell mass, computes area and coverage from frozen scores, joins point errors by kick and records plotted values and SHA-256 hashes. The scripts `coverage_benchmark.py` and `event_sensitivity.py` replay the 100 ms test and the crossing-perturbation trials, respectively. The reproduction guide locates their inputs, thresholds and checks.

For Z10, `report/recheck_spin.py` reconstructs 12 calibration thresholds, 7,848 held-out score rows and 56 grouped summaries from the replay archive. The post hoc threshold exchange uses `report/cross_spin_calibration.py` and the two replay archives.

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Report on the Use of AI

OpenAI Codex assisted with the physical analysis, numerical implementation, evidence checks, figure generation and revision of the English manuscript. This public draft contains the disclosure only. The complete original prompts and outputs are to be appended to the submission copy from the team's interaction records.